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An atom-photon quantum gate using an atomic clock qubit and a birrefringent cavity

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Thesis submitted within the M.Sc. Condensed Matter Physics

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February 16, 2026

Abstract

Single atoms coupled to optical cavities provide a powerful platform for photonic quantum information processing. Minimizing the influence of external factors like magnetic field fluctuations is a key goal in this endeavor. In this work, I investigate a novel gate protocol that employs a single ^{87}Rb atom prepared in a magnetic-field-insensitive clock qubit and coupled to a high-finesse birefringent Fabry–Perot cavity. The cavity's high birefringence creates two well-separated polarization eigenmodes, enabling polarization-selective reflectivity that depends on the atomic qubit state. This allows for the implementation of a controlled phase (CPHASE) gate between the atomic clock states and our photonic qubits. By rotating the basis of the photonic qubit, one is then able to realize an atom-photon controlled-NOT (CNOT) gate.

Show some results here and talk about how well it works and also how one might be able to use the gate

I present simulations based on input–output theory and master-equation modeling that quantify the conditional reflection amplitudes, gate truth table, and resulting fidelities under realistic experimental imperfections such as finite mode matching, cavity loss channels, and multiphoton contributions. Our results indicate that atom-state-dependent phase shifts on the photonic qubit are achievable in the current system, providing a viable path toward a robust, high-fidelity, cavity-assisted atom–photon quantum gate.

Contents

Chapter 1

Introduction

Bla bla bla quantum gate important for distant computation between two nodes very cool. Cite some stuff from kimble and duan here [1], talk about recent advancements or stuff that was similar(cite reiserer here [2]).

Basically the idea here is to have a good introduction into the topic. Here I can then talk about:
- what makes up quantum computation - why it's so difficult to scale up in a certain way (maybe look at different examples with large amounts of ^{87}Rb atoms and comment on what the challenges there are when it comes to first of all having lots of them, and them keeping them coherent and stuff like that). - Talk basically a lot about what Duan and Kimble talked about in their introduction of the paper since they were talking a lot about how computing at one node is hard, but spliting it up into smaller nodes and thus making it able to compute stuff. - Then talk about how one is then able to connect the nodes with the gate approach. - Explain what makes it unique (aka the birefringence of the cavity and the ability to then use the clock states which makes it insensitive to external magnetic fields) - Give an outlook over the chapters that are to follow

Chapter 2

Theory

2.1 Cavity Quantum Electrodynamics

2.1.1 Jaynes-Cummings Model

2.1.2 Dissipation and Decay Channels

2.1.2.1 Atomic Decay Channels

2.1.2.2 Photonic Decay Channels

2.2 Reflection from a Cavity-Atom System

2.2.1 Input-Output Formalism

2.2.2 Atom-State-Dependent Reflection

2.3 Conditional Reflection

2.3.1 Reflection in CNOT basis

Appendix A

Analytical Derivations

A.1 Analytical derivation for case of atom in $|0\rangle$ and photon is in $|h\rangle$

Our Hamiltonian is as follows:

$$H = \hbar g(|e\rangle\langle 1| a_h + |1\rangle\langle e| a_h^\dagger)$$

We drive the a_h mode of our cavity through $a_h^{in}(t)$, thus giving us the following equation of motion for our a_h cavity mode:

$$\dot{a_h} = -i[a_h, H] - (i\Delta + \frac{\kappa}{2})a_h - \sqrt{\kappa}a_h^{in}(t)$$

Because a_h commutes with H, the commutator is 0 and we get

$$\rightarrow \dot{a_h} = -(i\Delta + \frac{\kappa}{2})a_h - \sqrt{\kappa}a_h^{in}(t)$$

We now drop the subscript h as we'll only be looking at that case.
Multiplying with $e^{i\omega t}$, and integrating we get:

$$\int dte^{i\omega t}\dot{a}(t) = \underbrace{[e^{i\omega t}a(t)]_{-\infty}^{\infty}}_{=0} - \underbrace{i\omega \int dte^{i\omega t}a(t)}_{=i\omega a(\omega)}$$

where in the last expression we used that the integral performs a Fourier transformation.

$$\Rightarrow -i\omega a(\omega) = -(i\Delta + \frac{\kappa}{2})a(\omega) - \sqrt{\kappa}a^{in}(\omega)$$
$$\sqrt{\kappa}a^{in}(\omega) = -(i\Delta + \frac{\kappa}{2} - i\omega)a(\omega)$$
$$\rightarrow a(\omega) = -\frac{\sqrt{\kappa}}{i\Delta + \frac{\kappa}{2} - i\omega}a^{in}(\omega)$$

Now from our regular input output relation

$$a^{out}(\omega) = a^{in}(\omega) + \kappa a(\omega)$$

We thus get:

$$
\begin{aligned}
a^{out}(\omega) &= a^{in}(\omega) - \frac{\sqrt{\kappa}}{i\Delta + \frac{\kappa}{2} - i\omega} a^{in}(\omega) \\
a^{out}(\omega) &= (1 - \frac{\kappa}{i\Delta + \frac{\kappa}{2} - i\omega}) a^{in}(\omega) \\
a^{out}(\omega) &= (1 - \frac{i\Delta + \frac{\kappa}{2} - i\omega - \kappa}{i\Delta + \frac{\kappa}{2} - i\omega}) a^{in}(\omega) \\
a^{out}(\omega) &= (\frac{i\Delta - \frac{\kappa}{2} - i\omega}{i\Delta + \frac{\kappa}{2} - i\omega}) a^{in}(\omega) \\
&= \underbrace{\frac{i(\Delta-\omega) - \frac{\kappa}{2}}{i(\Delta-\omega) + \frac{\kappa}{2}}}_{h(\omega)} a^{in}(\omega)
\end{aligned}
$$

We can then use the convolution theorem in order to solve for $a^{out}(t)$

$$
a^{out}(t) = \int_{-\infty}^{\infty} d\tau \qquad \underbrace{h(t-\tau)}_{h(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega e^{-i\omega t} h(\omega)} \qquad a^{in}(\tau)
$$

Now using that $h(\omega) = 1 - \frac{\kappa}{i(\Delta-\omega)+\frac{\kappa}{2}}$ we get for $h(t)$ (via partial integration):

$$
\begin{aligned}
h(t) &= \delta(t) - \kappa e^{-i\delta t} e^{-\frac{\kappa}{2}t} \theta(t) \\
\Rightarrow a^{out}(t) &= a^{in}(t) - \kappa \int_{-\infty}^{\infty} d\tau e^{-i\delta(t-\tau)} e^{-\frac{\kappa}{2}(t-\tau)} a^{in}(t)
\end{aligned}
$$

For slowly varying pulses we can now approximate that $a^{in}(\tau) \approx a^{in}(t)$ thus giving us:

$$
\begin{aligned}
\Rightarrow a^{out}(t) &\approx a^{in}(t) - \frac{\kappa}{i\Delta + \frac{\kappa}{2}} a^{in}(t) \\
&= \frac{i\Delta - \frac{\kappa}{2}}{i\Delta + \frac{\kappa}{2}} a^{in}(t)
\end{aligned}
$$

Which for no detuning (so $\Delta = 0$) gives us:

$$
a^{out}(t) = -a^{in}(t) \tag{A.1}
$$

A.1.1 Lindblad Formalism

We start from the master equation:

$$
\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \sum_j \gamma_j \left(L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\} \right), \tag{A.2}
$$

Now, for an operator O,

$$
\begin{aligned}
\frac{d}{dt}\langle O \rangle &= \mathrm{Tr}\,(O\dot{\rho}) \qquad (A.3) \\
&= \mathrm{Tr}\left(O \left[-\frac{i}{\hbar}[H,\rho] + \sum_j \gamma_j \left(L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\} \right) \right] \right). \qquad (A.4)
\end{aligned}
$$

The Hamiltonian contribution gives

$$-\operatorname{Tr}\left(O\frac{i}{\hbar}[H,\rho]\right) = -\frac{i}{\hbar}\operatorname{Tr}\left(OH\rho - O\rho H\right) = -\frac{i}{\hbar}\operatorname{Tr}\left([O,H]\rho\right) = -\frac{i}{\hbar}\langle[O,H]\rangle. \tag{A.5}$$

For the dissipative terms we obtain

$$\sum_j \gamma_j \operatorname{Tr}\left(OL_j\rho L_j^\dagger - \tfrac{1}{2}OL_j^\dagger L_j\rho - \tfrac{1}{2}O\rho L_j^\dagger L_j\right). \tag{A.6}$$

Using cyclicity of the trace:

$$= \sum_j \gamma_j \left(\langle L_j^\dagger O L_j\rangle - \tfrac{1}{2}\langle\{L_j^\dagger L_j, O\}\rangle\right). \tag{A.7}$$

Therefore the general equation of motion is

$$\frac{d}{dt}\langle O\rangle = -\frac{i}{\hbar}\langle[O,H]\rangle + \sum_j \gamma_j \left(\langle L_j^\dagger O L_j\rangle - \tfrac{1}{2}\langle\{L_j^\dagger L_j, O\}\rangle\right). \tag{A.8}$$

A.1.2 Calculation of the Maxwell Bloch Equations (no lossy channels)

A.1.2.1 Jaynes Cummings Hamiltonian

Assuming ground state has zero energy our Jaynes-Cummings Hamiltonian is equal to:

$$H_{JC,S} = \underbrace{\omega_c a^\dagger a + \omega_a \sigma^\dagger \sigma}_{=H_{0,S}} + \underbrace{g(\sigma+\sigma^\dagger)(a+a^\dagger)}_{=H_{int,S}} + H_{drive,S} \tag{A.9}$$

where σ is the atomic lowering operator,ω_a the atomic transition frequency between the levels $|1\rangle$ and $|e\rangle$,a the photonic destruction operator, and ω_a the cavity mode resonance frequency.
H_{drive} in this case is the external light field which is driving our system, which in the semiclassical formalism has the following form [3]:

$$H_{drive,S} = i(\epsilon e^{-i\omega_d t} + \epsilon^* e^{i\omega_d t})(a+a^\dagger) \tag{A.10}$$

Here, a again is the photonic destruction operator, ω_d is the frequency of the driving field, and ϵ is the complex amplitude of our incoming pulse.
In general going from the Schrödinger picture to the interaction picture, is achieved by converting our operators into the interaction picture via following equation:

$$i\frac{dA_I}{dt} = [A_I(t), H_{0,S}] \tag{A.11}$$

Furthermore, $H_{0,I} = H_{0,S}$. Equipped with this we can now calculate the expressions for operators in the interaction picture:

$$\begin{aligned} i\frac{da_I(t)}{dt} &= [a_I(t), H_{0,S}] && \text{(A.12)}\\ &= [a_I(t), \omega_c a^\dagger a] && \text{(A.13)}\\ &= \omega_c a_I(t) && \text{(A.14)}\\ \Rightarrow \frac{da_I(t)}{dt} &= -i\omega_c a_I(t) \rightarrow a_I(t) = a e^{-i\omega_c t} && \text{(A.15)} \end{aligned}$$

Where we used that:

$$[a, H_{0,S}] = \omega_c[a, a^\dagger a] = \omega_c(\underbrace{[a,a^\dagger]}_{=1} a + a^\dagger \underbrace{[a,a]}_{=0}) = \omega_c a \tag{A.16}$$

Now we repeat the same thing for the atomic lowering operator σ:

$$i\frac{d\sigma_I(t)}{dt} = \underbrace{[\sigma_I(t), H_{0,S}]} \tag{A.17}$$

$$i\frac{d\sigma_I(t)}{dt} = \omega_a\sigma_I(t) \tag{A.18}$$

$$\Rightarrow \frac{d\sigma_I(t)}{dt} = -i\omega_a\sigma_I(t) \to \sigma_I(t) = \sigma e^{-i\omega_a t} \tag{A.19}$$

Now we can calculate the Hamiltonian for the interaction part and the drive part, in the interaction picture, by just plugging these operators in there, giving us:

$$H_{int,I} = \sigma a e^{-i(\omega_a-\omega_c)t} + \sigma a^\dagger e^{-i(\omega_a-\omega_c)t} + \sigma^\dagger a e^{-i(\omega_c-\omega_a)t} + \sigma^\dagger a^\dagger e^{i(\omega_c+\omega_a)t} \tag{A.20}$$

$$\xrightarrow{RWA} = \sigma a^\dagger e^{-i(\omega_a-\omega_c)t} + \sigma^\dagger a e^{-i(\omega_c-\omega_a)t} \tag{A.21}$$

$$H_{drive,I} = i(\epsilon e^{-i\omega_d t} + \epsilon^\dagger e^{i\omega_d t})(a_I(t) + a_I^\dagger(t)) \tag{A.22}$$

$$\xrightarrow{RWA} = i(a\epsilon^* e^{-i(\omega_c-\omega_d)t} + a^\dagger\epsilon e^{i(\omega_c-\omega_d)t}) \tag{A.23}$$

Thus giving us the total Hamiltonian in the interaction picture under the rotating wave approximation:

$$H_{JC,I} \overset{\text{RWA}}{=} \omega_c a^\dagger a + \omega_a\sigma^\dagger\sigma + g(\sigma a^\dagger e^{-i(\omega_a-\omega_c)t} + \sigma^\dagger a e^{-i(\omega_c-\omega_a)t}) + i(a\epsilon^* e^{-i(\omega_c-\omega_d)t} + a^\dagger\epsilon e^{i(\omega_c-\omega_d)t}) \tag{A.24}$$

If we now transform back into the Schrödinger picture, we get:

$$H_{JC,S} \overset{\text{RWA}}{=} \omega_c a^\dagger a + \omega_a\sigma^\dagger\sigma + g(a^\dagger\sigma + a\sigma^\dagger) + i(a\epsilon^* e^{i\omega_d t} + a^\dagger\epsilon e^{-i\omega_d t}) \tag{A.25}$$

If we now change into the rotating frame of the driving field via

$$U(t) = \exp\{i\omega_d t(a^\dagger a + \sigma^\dagger\sigma)\} \tag{A.26}$$

$$H \to UHU^\dagger + i\dot{U}U^\dagger =: H_{RF} \tag{A.27}$$

we then get the final form of our Hamiltonian:

$$\boxed{H_{JC,RF} = \underbrace{(\omega_c - \omega_d)}_{=\Delta_c} a^\dagger a + \underbrace{(\omega_a - \omega_d)}_{=\Delta_a} \sigma^\dagger\sigma + g(a^\dagger\sigma + a\sigma^\dagger) + i(a\epsilon^* + a^\dagger\epsilon)} \tag{A.28}$$

Which gives rise to 4 scenarios which I'll be covering with my project:

scenario	**meaning**
$\Delta_a = \Delta_c = 0$	Atom is in coupled state (here: $\|1\rangle$ and we drive with a light field that's resonant to the cavity (here: we drive $\|1\rangle \leftrightarrow \|e\rangle$ which would be our $\|\pi\rangle$ polarized light
$\Delta_a = 0; \Delta_c \neq 0$	Atom is in coupled state but we drive with light that's non-resonant to cavity (here: $\|V\rangle$ polarized light)
$\Delta_a \neq 0; \Delta_c = 0$	Atom is in uncoupled state (here: $\|0\rangle$) and we drive with a light field that's resonant to the cavity
$\Delta_a, \Delta_c \neq 0$	Atom is in uncoupled state and we drive with light that's non-resonant to cavity

In order to improve readability we'll be splitting the full JC Hamiltonian into three parts:

$$H_0 = \Delta_c a^\dagger a + \Delta_a\sigma^\dagger\sigma \tag{A.29}$$

$$H_{int} = g(a^\dagger\sigma + a\sigma^\dagger) \tag{A.30}$$

$$H_{drive} = i(a\epsilon^* + a^\dagger\epsilon) \tag{A.31}$$

with following losses:

$$\gamma_1 = \gamma, \quad L_1 = \sigma, \tag{A.32}$$

$$\gamma_2 = \kappa, \quad L_2 = a. \tag{A.33}$$

A.1.3 Cavity mode a

For H_0:

$$[a, H_0] = \Delta_c[a, a^\dagger a] + \Delta_c a \underbrace{[a, \sigma^\dagger \sigma]}_{=0} \tag{A.34}$$

$$= \Delta_c(\underbrace{[a, a^\dagger]}_{=1} a + a^\dagger \underbrace{[a, a]}_{=0}) \tag{A.35}$$

$$= \Delta_c a \tag{A.36}$$

$$\boxed{\Rightarrow -i\langle [a, H_0] \rangle = -i\Delta_c \langle a \rangle} \tag{A.37}$$

For H_{int}:

$$[a, H_{int}] = g([a, a^\dagger \sigma] + [a, a\sigma^\dagger]) \tag{A.38}$$

$$= g(aa^\dagger \sigma - a^\dagger a\sigma + \cancel{aa\sigma^\dagger} - \cancel{aa\sigma^\dagger}) \tag{A.39}$$

$$\rightarrow \langle aa^\dagger \sigma \rangle = \langle n| aa^\dagger \sigma |n\rangle = (n+1)\langle \sigma \rangle \tag{A.40}$$

$$\rightarrow \langle a^\dagger a\sigma \rangle = \langle n| a^\dagger a\sigma |n\rangle = n\langle \sigma \rangle \tag{A.41}$$

$$\boxed{\Rightarrow -i\langle [a, H_{int}] \rangle = -ig\langle \sigma \rangle} \tag{A.42}$$

For H_{drive}:

$$[a, H_{drive}] = [a, i(a\epsilon^* + a^\dagger \epsilon)] \tag{A.43}$$

$$= i([a, a\epsilon^*] + [a, a^\dagger \epsilon]) \tag{A.44}$$

$$= i(0 + \epsilon) \tag{A.45}$$

$$= i\epsilon \tag{A.46}$$

$$\boxed{\Rightarrow -i\langle [a, H_{drive}] \rangle = i\langle \epsilon \rangle \overset{\epsilon \in \mathbb{C}}{=} i\epsilon} \tag{A.47}$$

If we now include the fact that:

$$i\epsilon \mathrel{\widehat{=}} \sqrt{\kappa_{oc}} a^{in}(t) \tag{A.48}$$

we are then left with:

$$\boxed{\Rightarrow -i\langle [a, H_{drive}] \rangle = -\sqrt{\kappa_{oc}} a^{in}(t)} \tag{A.49}$$

For rest of the Lindbladian:

$$D(a) = a^\dagger aa - \frac{1}{2}\{a^\dagger a, a\} \tag{A.50}$$

$$= a^\dagger aa - \frac{1}{2}(a^\dagger aa - aa^\dagger a) \tag{A.51}$$

$$= \frac{1}{2}a^\dagger aa - \frac{1}{2}aa^\dagger a \tag{A.52}$$

$$= \frac{1}{2}(a^\dagger aa - aa^\dagger a) \tag{A.53}$$

$$= \frac{1}{2}([a^\dagger, a]a) \tag{A.54}$$

$$= \frac{1}{2}(-a) \tag{A.55}$$

$$= -\frac{1}{2}a \tag{A.56}$$

$$\boxed{\Rightarrow \kappa\langle D(a) \rangle = -\frac{\kappa}{2}\langle a \rangle} \tag{A.57}$$

So for $\langle a\rangle(t)$ we get:

$$\boxed{\begin{aligned}\frac{d\langle a\rangle}{dt} &= -i\Delta_c\langle a\rangle - ig\langle\sigma\rangle - \sqrt{\kappa_{oc}}\,a_{\text{in}}(t) - \frac{\kappa}{2}\langle a\rangle \\ &= -(i\Delta_c + \tfrac{\kappa}{2})\langle a\rangle - ig\langle\sigma\rangle - \sqrt{\kappa_{oc}}\,a_{\text{in}}(t)\end{aligned}} \tag{A.58}$$

A.1.4 Atomic lowering operator σ

For H_0:

$$[\sigma, H_0] = \Delta_a \underbrace{[\sigma, \sigma^\dagger\sigma]}_{=\underbrace{[\sigma,\sigma^\dagger]}_{=\mathbb{1}}\sigma + \sigma^\dagger\underbrace{[\sigma,\sigma]}_{=0}} \tag{A.59}$$

$$= \Delta_a \mathbb{1}\sigma \tag{A.60}$$

$$\boxed{\Rightarrow -i\langle[\sigma, H_0]\rangle = -i\Delta_a\langle\sigma\rangle} \tag{A.61}$$

For H_{int}:

$$[\sigma, H_{int}] = g([\sigma, a^\dagger\sigma] + [\sigma, a\sigma^\dagger]) \tag{A.62}$$

$$= g(\cancel{a^\dagger\sigma\sigma} - \cancel{a^\dagger\sigma\sigma} + a\sigma\sigma^\dagger - a\sigma^\dagger\sigma) \tag{A.63}$$

$$= g(a\sigma\sigma^\dagger - a\sigma^\dagger\sigma) \tag{A.64}$$

Now we can consider that:

$$\boxed{\begin{aligned}\sigma^\dagger\sigma &= |e\rangle\langle 1\rangle\langle e| = |e\rangle\langle e| \\ \sigma\sigma^\dagger &= |1\rangle\langle e\rangle\langle 1| = |1\rangle\langle 1|\end{aligned}} \tag{A.65, A.66}$$

$$[\sigma, H_{int}] = g(a\sigma\sigma^\dagger - a\sigma^\dagger\sigma) \tag{A.67}$$

$$= ga(|1\rangle\langle 1| - |e\rangle\langle e|) \tag{A.68}$$

$$= ga\sigma_z \tag{A.69}$$

$$\boxed{\Rightarrow -i\langle[\sigma, H_{int}]\rangle = -ig\langle a\sigma_z\rangle} \tag{A.70}$$

No interaction with H_{drive}.
For rest of Lindbladian:

$$D(\sigma) = \underbrace{\sigma^\dagger\underbrace{\sigma\sigma}_{=0}}_{=0} - \frac{1}{2}\{\sigma^\dagger\sigma, \sigma\} \tag{A.71}$$

$$= -\frac{1}{2}\sigma^\dagger\underbrace{\sigma\sigma}_{=0} - \frac{1}{2}\sigma\sigma^\dagger\sigma \tag{A.72}$$

$$= -\frac{1}{2}\sigma \tag{A.73}$$

$$\boxed{\gamma\langle D(\sigma)\rangle = -\frac{\gamma}{2}\langle\sigma\rangle} \tag{A.74}$$

So for $\langle\sigma\rangle$(t) we get:

$$\boxed{\begin{aligned}\frac{d\langle\sigma\rangle}{dt} &= -i\Delta_a\langle\sigma\rangle - ig\langle a\sigma_z\rangle - \frac{\gamma}{2}\langle\sigma\rangle \\ &= -(i\Delta_a + \frac{\gamma}{2})\langle\sigma\rangle - ig\langle a\sigma_z\rangle\end{aligned}} \tag{A.75}$$

A.1.5 Pauli-Z operator σ_z

For H_0:

$$\begin{aligned}
[\sigma_z, H_0] &= \Delta_a[\sigma_z, \sigma^\dagger\sigma] &\text{(A.76)}\\
&= \Delta_a[\sigma_z, |e\rangle\langle e|] &\text{(A.77)}\\
&= \Delta_a(\sigma_z|e\rangle\langle e| - |e\rangle\langle e|\sigma_z) &\text{(A.78)}\\
&= \Delta_a(-|e\rangle\langle e| + |e\rangle\langle e|) &\text{(A.79)}\\
&= 0 &\text{(A.80)}
\end{aligned}$$

For H_{int}:

$$\begin{aligned}
[\sigma_z, H_{int}] &= g([\sigma_z, a^\dagger\sigma] + [\sigma_z, a\sigma^\dagger]) &\text{(A.81)}\\
&= g(a^\dagger\sigma_z\sigma - a^\dagger\sigma\sigma_z + a\sigma_z\sigma^\dagger - a\sigma^\dagger\sigma_z) &\text{(A.82)}\\
&= g(a^\dagger\sigma + a^\dagger\sigma - a\sigma^\dagger - a\sigma^\dagger) &\text{(A.83)}\\
&= 2g(a^\dagger\sigma - a\sigma^\dagger) &\text{(A.84)}
\end{aligned}$$

$$\boxed{\Rightarrow -i\langle[\sigma_z, H_{int}]\rangle = -2ig(\langle a^\dagger\sigma\rangle - \langle a\sigma^\dagger\rangle)} \tag{A.85}$$

No interaction with H_{drive}. For rest of Lindbladian:

$$\begin{aligned}
D(\sigma_z) &= \sigma^\dagger\sigma_z\sigma - \frac{1}{2}\{\sigma^\dagger\sigma, \sigma\} &\text{(A.86)}\\
&= |e\rangle\langle e| - \frac{1}{2}\underbrace{\sigma^\dagger\sigma\sigma_z}_{=-|e\rangle\langle e|} - \frac{1}{2}\underbrace{\sigma_z\sigma^\dagger\sigma}_{=-|e\rangle\langle e|} &\text{(A.87)}\\
&= |e\rangle\langle e| + |e\rangle\langle e| &\text{(A.88)}\\
&= 2|e\rangle\langle e| &\text{(A.89)}\\
&= \mathbb{1} - \sigma_z &\text{(A.90)}
\end{aligned}$$

$$\boxed{\Rightarrow \gamma\langle D(\sigma_z)\rangle = \gamma(1 - \langle\sigma_z\rangle)} \tag{A.91}$$

So for $\langle\sigma_z\rangle(t)$ we get:

$$\boxed{\frac{d\langle\sigma_z\rangle(t)}{dt} = -2ig(\langle a^\dagger\sigma\rangle - \langle a\sigma^\dagger\rangle) + \gamma(1 - \langle\sigma_z\rangle)} \tag{A.92}$$

The final Maxwell Bloch Equations then take the form of:

$$\boxed{\begin{aligned}
\frac{d\langle a\rangle}{dt} &= -(i\Delta_c + \tfrac{\kappa}{2})\langle a\rangle - ig\langle\sigma\rangle - \sqrt{\kappa_{oc}}\, a_{\text{in}}(t) &\text{(A.93)}\\
\frac{d\langle\sigma\rangle}{dt} &= -(i\Delta_a + \tfrac{\gamma}{2})\langle\sigma\rangle - ig\langle a\sigma_z\rangle &\text{(A.94)}\\
\frac{d\langle\sigma_z\rangle}{dt} &= -2ig\big(\langle a^\dagger\sigma\rangle - \langle a\sigma^\dagger\rangle\big) + \gamma\big(1 - \langle\sigma_z\rangle\big) &\text{(A.95)}
\end{aligned}}$$

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Selbständigkeitserklärung

Ich versichere hiermit, dass ich die von mir eingereichte Abschlussarbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

gezeichnet,
Leart Zuka, München den TBD.